Ng Yu Hang (Mervin)

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EDUCATION

Singapore Polytechnic Singapore

Diploma in Applied AI & Analytics; CGPA 3.83 / 4.00 *Apr 2024 - Apr 2027 (expected)*

- Director's Honour Roll AY25/26; top 5% in Deep Learning, Practical AI, DevOps for AI

EXPERIENCE

Founding Product & AI Engineer (Part-time) Sep 2026 - Present

Influencees *Singapore (Hybrid)*

- Built Ai-kyo, an agentic AI assistant for creators, by orchestrating an intent-routed LLM agent with tool integrations
- Unified 6 messaging channels into one inbox (WhatsApp, Instagram, LinkedIn, TikTok, Gmail and more)
- Cut inbox sign-in lag from 1.1 s to 12 ms by profiling the streaming path; shipped 10 PRs to production

Full-Stack AI Engineer (Intern) Mar 2026 - Present

SP Data Science & Analytics Centre *Singapore*

- Cut retrieval time ~70-85% and manual search ~60% for EM Services with an Azure AI Search RAG chatbot
- Cut Power BI grading time ~50-70% with an LLM grader that scores .pbix dashboards against rubrics

AI Researcher / Developer (Part-time) May 2026 - Aug 2026

PetaniAI *Singapore (Hybrid)*

- Researched and tested emerging AI models, tools and technologies, prototyping the most promising
- Informed model choices by benchmarking LLMs on the company's own requirements

Student Developer Sep 2025 - Feb 2026

Singapore Polytechnic, Department of Communications *Singapore*

- Built the full stack of a RAG chat agent on AWS for SP Communications (paid project)
- Owned the agentic layer: guardrails and LLM-as-a-judge evaluation; presented to SP's Deputy Principal and PCEO

PERSONAL PROJECTS

AgentLabs | *Python, FastAPI, React, PyTorch (LoRA), Docker* 2026

- Hardened AI agents against 200 adversarial attacks via an attack, verify, auto-patch and retest loop (live demo)

ProofBench | *Python, FastAPI, PostgreSQL, Docker, GitHub Actions* 2026

- Evaluated developer tools by generating adapters, running them in sandboxes and scoring output deterministically

Lactic | *Python, Flutter, Supabase, embeddings* 2026

- Built an A-Level AI tutor with hybrid vector and graph retrieval over 31k+ chunks

Edge Human Detection | *Python, PyTorch, YOLO, OpenCV, Docker* 2026

- Fine-tuned YOLO11 on thermal imagery for night-vision person detection with multi-frame checks

HONORS & AWARDS

- **National Olympiad in Artificial Intelligence (NOAI) 2026**: Bronze, top 60 in all of Singapore
- **Projet: Hack & Hire**: Winner
- **OpenClaw Hackathon**: Top 3
- **Agent Forge Hackathon**: 3rd runner-up
- **PolyFinTech 2026 (ESG Category)**: Finalist
- **'SUP Build2026**: Top 3, Best Use of Zo
- **Daytona HackSprint**: 2nd runner-up
- **NAISC (Huawei Track)**: Top 10

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, HTML, CSS, Dart
Generative AI	RAG, agents, tool use, LLM evaluation, LoRA fine-tuning, prompt engineering, VLMs
ML	PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, YOLO, OpenCV
Engineering	FastAPI, Node.js, React, Next.js, Docker, Azure, AWS, PostgreSQL, Supabase, GitHub Actions
Certifications	AWS Certified AI Practitioner, Google AI Professional, AI Singapore Certified Trainer